Complex Numbers

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

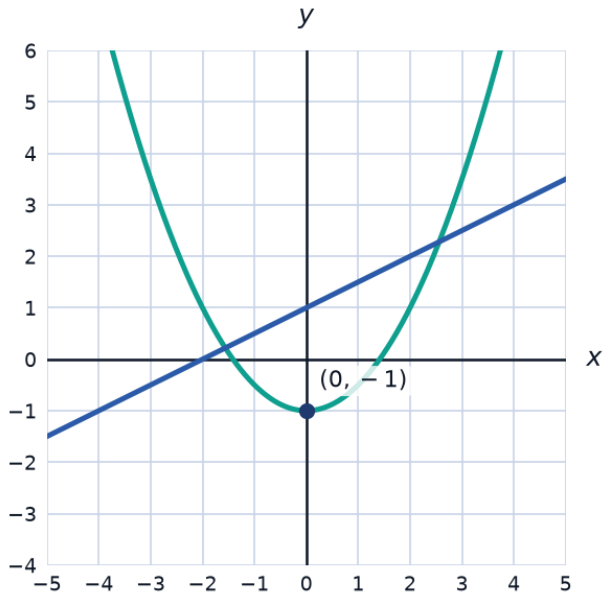
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Arithmetic with complex numbers

Let $z_1 = 4 - 2i$ and $z_2 = -3 + i$. Find $z_1 + z_2$ and $z_1 z_2$.

2 [Foundational] • Target: Arithmetic with complex numbers

Find $(6 + i) - (2 - 5i)$.



3 [Foundational] • Target: Complex roots of a quadratic

Solve $x^2 + 6x + 13 = 0$.

4 [Foundational] • Target: Complex roots of a quadratic

Solve $2x^2 - 4x + 10 = 0$.

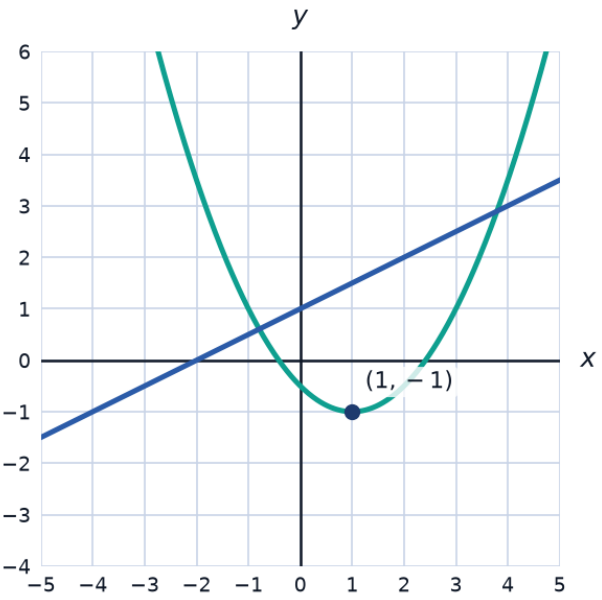
5 [Foundational] • Target: Complex roots of a quadratic

Without solving it, explain how you know $x^2 + 2x + 5 = 0$ has complex (not real) roots.

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6 [Foundational] • Target: Complex roots of a quadratic

Verify that $x = -3 + 2i$ is a root of $x^2 + 6x + 13 = 0$ by direct substitution.



7 [Foundational] • Target: Modulus

Find $|z|$ for $z = -3 + 2i$ (one of the roots found above).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Modulus

Find $|z|$ for $z = 5 - 12i$.

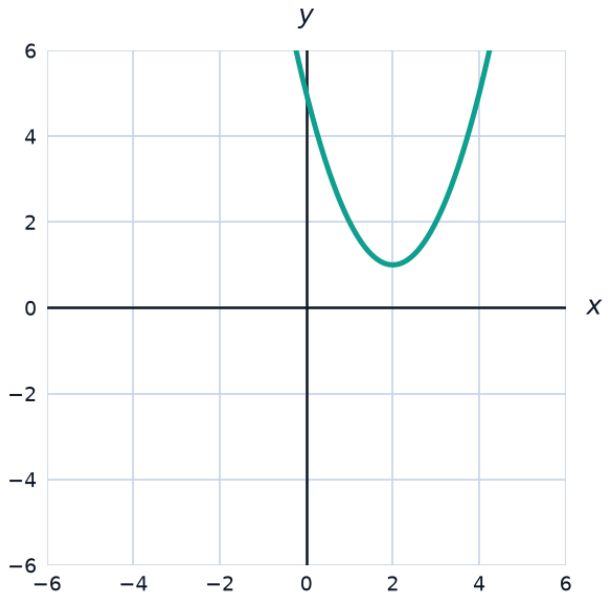
9 [Intermediate] • Target: Modulus

Explain why a conjugate pair $a \pm bi$ always has the same modulus for both roots.

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10 **[Intermediate]** • **Target: Modulus**

Let z_1 and z_2 be the roots of the quadratic equation $x^2 - 4x + 5 = 0$. Consider a complex number w such that $w = z_1^2 + z_2^2$. If w is expressed in the form $a + bi$, where a and b are real numbers, find the value of $a - b$.



11 **[Intermediate]** • **Target: Modulus-Argument Form**

Let the complex number w be defined as $w = (a + bi) + (3 - 2i)$, where a and b are real numbers. Given that $|w|^2 = 25$ and w lies in the first quadrant, find the value of $a + b$.

12 **[Intermediate]** • **Target: De Moivre's Powers**

Let $z_1 = 2 + 3i$ and $z_2 = 1 - i$. Consider the complex number w such that $w = \frac{z_1^2}{z_2}$. If $w = a + bi$, where a and b are real numbers, find the value of b.

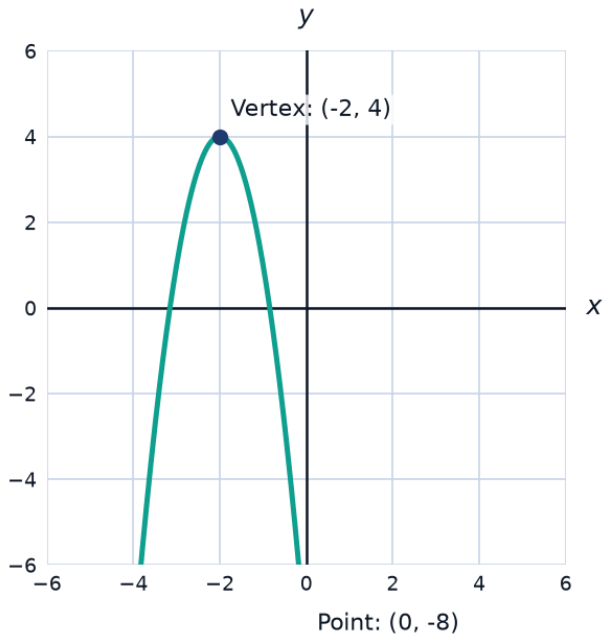
13 **[Intermediate]** • **Target: Roots of Complex Numbers**

Consider the complex number $z = 3 + 4i$. What is the modulus of z?

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14 [Intermediate] • Target: Complex Geometry & Loci

Consider the function $g(x) = ax^2 + bx + c$. The graph of this quadratic function is shown below. The vertex of the parabola is at $(-2, 4)$. The parabola passes through the point $(0, -8)$. Find the value of the parameter a.

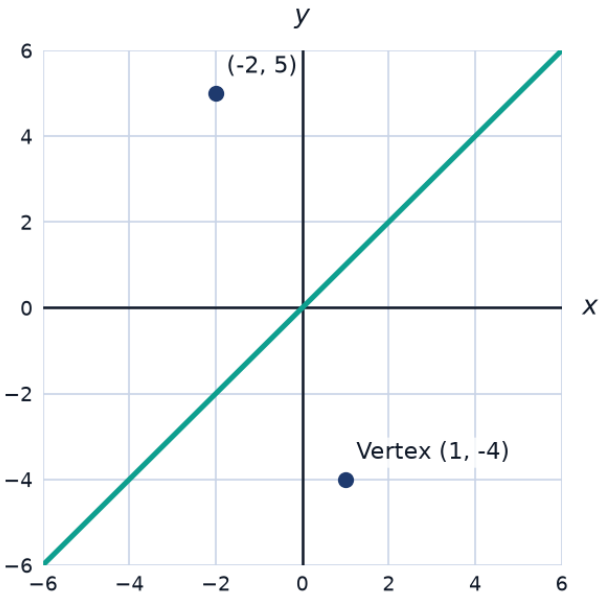


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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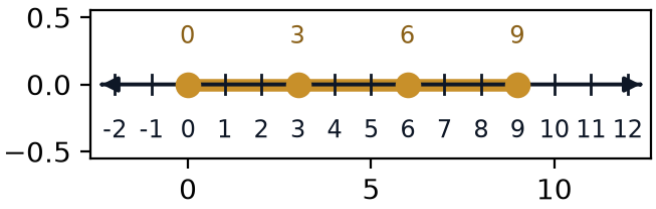
15 [Challenging] • Target: Arithmetic with complex numbers • Cross-Lesson: 6.01 Probability

The graph of the quadratic function $f(x) = x^2 + bx + c$ is shown below. The graph passes through the point $(-2, 5)$ and has its vertex at $(1, -4)$. Determine the value of b.



16 [Challenging] • Target: Complex roots of a quadratic • Cross-Lesson: 2.01 Sequences & Series

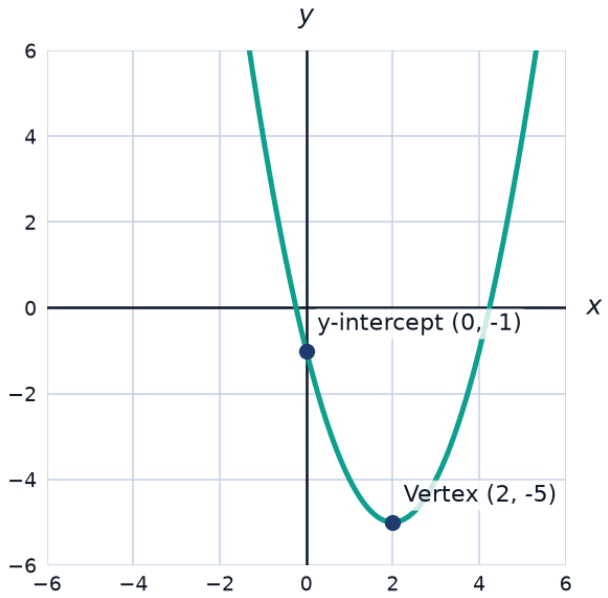
Consider the arithmetic sequence shown on the number line. What is the common difference of this sequence?



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17 [Challenging] • Target: Modulus • Cross-Lesson: 3.01 Functions & Graphs

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at $(2, -5)$ and the y-intercept is at $(0, -1)$. Determine the value of a.



18 [Challenging] • Target: Modulus-Argument Form • Cross-Lesson: 4.01 Matrices & Vectors

Consider an arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the value of the 4th term, a_4?

19 [Challenging] • Target: De Moivre's Powers • Cross-Lesson: 5.01 Introductory Calculus

An arithmetic sequence has its first term $a_1 = 7$ and a common difference $d = 5$. What is the value of the 3rd term, a_3?

20 [Challenging] • Target: Roots of Complex Numbers • Cross-Lesson: 6.01 Probability

Consider the complex number $z = 3 + 4i$. What is the magnitude of z?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Arithmetic with complex numbers

Let $z_1 = -6 + 9i$ and $z_2 = 2 + 6i$. Find $z_1 + z_2$.

a. $-2 + 15i$

b. $-4 + 13i$

c. $-4 + 15i$

d. $-8 + 3i$

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22 [Foundational] • Target: Complex roots of a quadratic

Let $z_1 = 3 + 8i$ and $z_2 = -9 - 7i$. Find $z_1 + z_2$.

a. $12 + 15i$

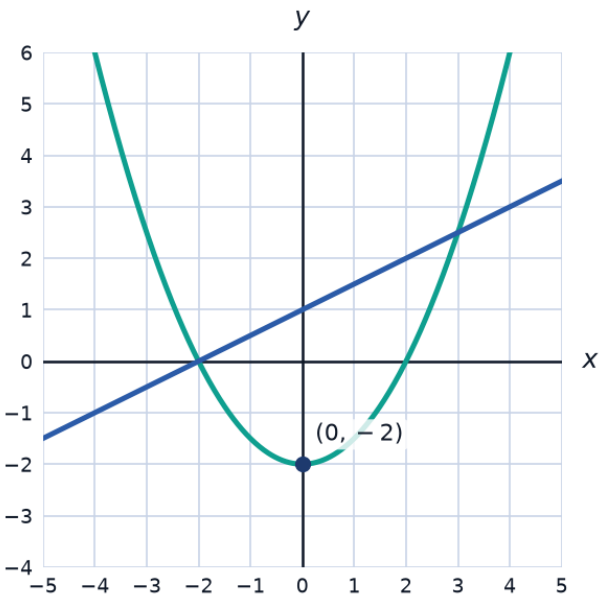
b. $-6 - i$

c. $-4 + i$

d. $-6 + i$

23 [Foundational] • Target: Modulus

Let $z_1 = -8 - 8i$ and $z_2 = 5 + 3i$. Find $z_1 + z_2$.



a. $-13 - 11i$

b. $-3 - 5i$

c. $-1 - 5i$

d. $-3 - 7i$

24 [Foundational] • Target: Modulus-Argument Form

Let $z_1 = -7 - 3i$ and $z_2 = -4 + 2i$. Find $z_1 + z_2$.

a. $-3 - 5i$

b. $-11 - 3i$

c. $-9 - i$

d. $-11 - i$

25 [Foundational] • Target: De Moivre's Powers

Let $z_1 = 5 - i$ and $z_2 = -3 - 6i$. Find $z_1 + z_2$.

a. $2 - 9i$

b. $8 + 5i$

c. $2 - 7i$

d. $4 - 7i$

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26 [Foundational] • Target: Roots of Complex Numbers

Let $z_1 = 5 + 2i$ and $z_2 = 8 - 2i$. Find $z_1 + z_2$.

a. 13

b. $-3 + 4i$

c. $13 - 2i$

d. 15

27 [Foundational] • Target: Complex Geometry & Loci

Let $z_1 = -8 - i$ and $z_2 = -5 + 6i$. Find $z_1 + z_2$.

a. $-13 + 3i$

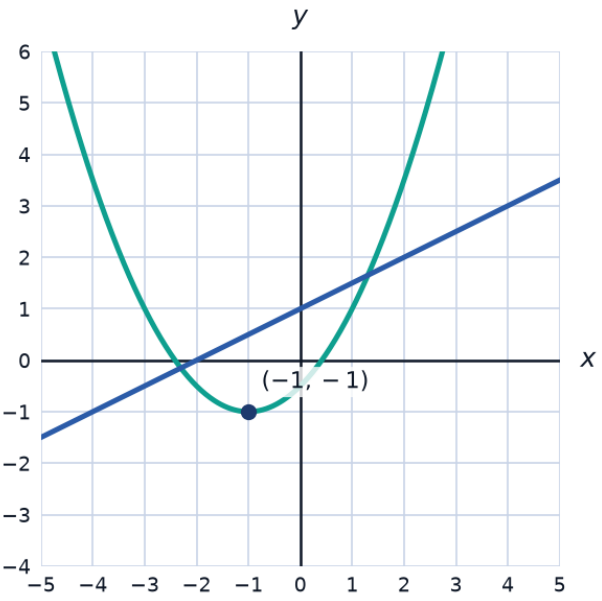
b. $-3 - 7i$

c. $-11 + 5i$

d. $-13 + 5i$

28 [Foundational] • Target: Arithmetic with complex numbers

Solve $x^2 - 2x + 5 = 0$.



a. $1 \pm 2i$

b. $1 \pm 3i$

c. $2 \pm 2i$

d. $-1 \pm 2i$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Complex roots of a quadratic

Solve $x^2 + 12x + 72 = 0$.

a. $6 \pm 6i$

b. $-6 \pm 6i$

c. $-6 \pm 7i$

d. $-5 \pm 6i$

30 [Intermediate] • Target: Modulus

Solve $x^2 - 6x + 58 = 0$.

a. $3 \pm 7i$

b. $4 \pm 7i$

c. $-3 \pm 7i$

d. $3 \pm 8i$

31 [Intermediate] • Target: Modulus-Argument Form

Solve $x^2 - 10x + 41 = 0$.

a. $5 \pm 4i$

b. $-5 \pm 4i$

c. $5 \pm 5i$

d. $6 \pm 4i$

32 [Intermediate] • Target: De Moivre's Powers

Solve $x^2 - 6x + 45 = 0$.

a. $3 \pm 7i$

b. $4 \pm 6i$

c. $-3 \pm 6i$

d. $3 \pm 6i$

33 [Intermediate] • Target: Roots of Complex Numbers

Solve $x^2 - 16x + 113 = 0$.

a. $9 \pm 7i$

b. $8 \pm 8i$

c. $-8 \pm 7i$

d. $8 \pm 7i$

34 [Intermediate] • Target: Complex Geometry & Loci

Solve $x^2 + 16x + 80 = 0$.

a. $8 \pm 4i$

b. $-7 \pm 4i$

c. $-8 \pm 4i$

d. $-8 \pm 5i$

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35 [Intermediate] • Target: Arithmetic with complex numbers

Find $|z|$ for $z = -1 + 6i$.

a. 37

b. $\sqrt{39}$

c. $\sqrt{35}$

d. $\sqrt{37}$

36 [Intermediate] • Target: Complex roots of a quadratic

Find $|z|$ for $z = -12 - 35i$.

a. 47

b. 39

c. 35

d. 37

37 [Intermediate] • Target: Modulus

Find $(2 - 7i) - (-6 - 9i) + (-8 - 5i)$.

a. $0 - 5i$

b. $3 - 3i$

c. $0 - 3i$

d. $2 - 3i$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Modulus-Argument Form • Cross-Lesson: 4.01 Matrices & Vectors

Find $(8 + i) - (5 - 3i) + (-8 + 9i)$.

a. $-5 + 11i$

b. $-5 + 13i$

c. $-3 + 13i$

d. $5 + 13i$

39 [Challenging] • Target: De Moivre's Powers • Cross-Lesson: 5.01 Introductory Calculus

Multiply $(3 - i)(6 + 7i)$.

a. $25 + 27i$

b. $25 + 15i$

c. $25 + 17i$

d. $11 + 15i$

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40 [Challenging] • Target: Roots of Complex Numbers • Cross-Lesson: 6.01 Probability

Multiply $(6-4i)(-9+i)$.

a. $-50+44i$

b. $-50-30i$

c. $-50+42i$

d. $-58+42i$

41 [Challenging] • Target: Modulus-Argument Form • Cross-Lesson: 2.01 Sequences & Series

Find the polar representation of $z=-1-i\sqrt{3}$.

a. $2(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3})$

b. $2(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3})$

c. $2(\cos\frac{5\pi}{3}+i\sin\frac{5\pi}{3})$

d. $4(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3})$

42 [Challenging] • Target: Modulus-Argument Form • Cross-Lesson: 3.01 Functions & Graphs

If $z=4(\cos 70^\circ+i\sin 70^\circ)$ and $w=2(\cos 40^\circ+i\sin 40^\circ)$, what is $\frac{z}{w}$?

a. $2(\cos 30^\circ+i\sin 30^\circ)$

b. $2(\cos 110^\circ+i\sin 110^\circ)$

c. $8(\cos 30^\circ+i\sin 30^\circ)$

d. $2(\cos 2800^\circ+i\sin 2800^\circ)$

43 [Challenging] • Target: De Moivre's Powers • Cross-Lesson: 4.01 Matrices & Vectors

If $z+\frac{1}{z}=2\cos\theta$, what is $z^n+\frac{1}{z^n}$?

a. $2\cos(n\theta)$

b. $2\sin(n\theta)$

c. $\cos(n\theta)$

d. $2^n\cos^n\theta$

44 [Challenging] • Target: Roots of Complex Numbers • Cross-Lesson: 5.01 Introductory Calculus

The four 4th roots of -1 form the vertices of what geometric polygon in the complex plane?

a. A square

b. A rectangle

c. A rhombus

d. A regular hexagon

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45 [Challenging] • Target: Complex Geometry & Loci • Cross-Lesson: 6.01 Probability

Describe the locus of points in the complex plane satisfying $|z - 3| = |z - 5i|$.

a. Perpendicular bisector of segment joining $(3, 0)$ and $(0, 5)$

b. Circle of radius 3

c. Ellipse with foci at 3 and $5i$

d. Ray starting at 3

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